

Automated Analysis of Short Answers Using Text Vectorization

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Abstract— Open-ended questions are used to test knowledge, and automated assessment of answers to them is currently widely studied. The paper analyzes student's answers formulated in free form. At first, language models were used and assessments were obtained using cosine similarity. To clarify the assessments, it is proposed to use binary classification. The capabilities of explainable artificial intelligence tools and modern chatbots based on large language models were also studied.

Keywords— *knowledge test; computer-based assessment; BERT; LLM; cosine similarity; Automatic Short Answer Grading (ASAG); natural language processing; automated scoring; semantic similarity*

I. INTRODUCTION

When testing knowledge, questions that require a free-form answer from the student (open-ended questions) allow to increase the objectivity of the assessment. They help to obtain feedback on students' understanding of the material covered. Such questions should be widely used in tests, assignments, projects, so that the teacher can analyze and improve teaching strategies [1]. The open-ended question tool is an effective assessment tool for measuring the critical thinking skills of students of different ages, the learning level, the characteristics cognitive development of students [2]. Although the use of open-ended questions potentially yields richer information, it creates greater challenges in terms of extracting useful information, while significantly increasing the analyzes time. Since in most cases the assessment of answers is carried out by the teacher. [3]. Therefore, automatic evaluation of short answers is an important task in computer-aided assessment. Word embeddings, cosine similarity measure, and neural network-based classification models can be used to evaluate the similarity of students' answers [4]. The following paper presents a study of the problem of automatic classification of short coherent texts (essays) in English into the levels of the international CEFR scale. Determining the level of a text in natural language is an important component of student knowledge assessment. To solve this problem, vector models of text based on stylometric numerical features are considered. The resulting vectors are classified by standard machine learning classifiers [5]. Open-ended questions that require multi-label classification, to which multiple codes are assigned, are often encountered. The paper focuses on multi-label classification of text answers to open-ended questions. The authors evaluate the performance of the transformer-based BERT architecture for the German language compared to traditional multi-label algorithms [6]. The following paper presents a new hybrid BERT-based model. A customized mechanism adapted from BERT is used, which allows for reliably identification of semantic dependencies between

words in a sentence and, therefore, contributes to the efficiency and reliability of the scoring system [7]. The performance of various BERT-based models for aspect term extraction for Arabic texts obtained from educational institution surveys is studied. A system has been developed that automates the process of Sentiment Analysis based on aspects [8]. A popular application is to quantify semantic similarity between multidimensional objects by applying cosine similarity. But the authors warn of potential errors. They analytically deduced how cosine similarity can produce arbitrary "similarities" [9]. Binary classification deals with determining whether items belong to one of two possible categories. There are various metrics for evaluating the performance of such classification systems. In this paper, the authors presented a new mathematical approach to accurately evaluate binary classification metrics [10]. Although there has been a recent explosion of work in explainable artificial intelligence (XAI) on deep models that work with images and tabular data, text datasets present new challenges for the XAI community. Such challenges can be attributed to the lack of input structure in text data, the use of word embeddings, which increase the opacity of models and the difficulty visualizing the inner workings of deep models when they are trained on text data. Recently, methods have been developed to address the above issues and provide satisfactory explanations for natural language processing (NLP) models [11]. XAI methods such as LIME demonstrate the workings of a black-box model such as a random forest classifier. Using LIME improves the accuracy of sentiment analysis. Thus, explainability helps to better understand the model and its can be used to improve model performance, which in turn increases the reliability and validity of AI systems [12].

Thus, specialists are studying approaches to automated analysis of students' answers formulated in free form. Previously, we presented the results of evaluating answers to open-ended questions using language models [13]. In this paper, we will present the possibilities of refining the obtained grades.

II. DATA

We consider a database consisting of 1580 answers received from students. The answers were received during class; students did not have access to the Internet or reference books. There were 42 questions on topics related to computer graphics. Earlier we conducted an experiment, during which students were given an opportunity to repeat the theoretical material before the test. In this case, both the number of correct answers and the share of correctly evaluated answers increased. Since the students mainly use the words that were in the text for repetition [14].

After analyzing the answers, we concluded that the wording of some questions needed to be adjusted. We noticed that some students did not understand the task clearly enough. During automated checking, students and teachers do not have the opportunity to ask a clarifying question, and this must be taken into account when formulating questions. In this paper, we will consider the questions that were clearly understood by students. For example, the question “Give examples of gradational transformations of images” could be refined as “Give all examples of gradational transformations of images studied during the course” or “Give any two examples of gradational transformations of images”. Guidelines for instructors have been developed to address all questions raised by the students.

All answers were given expert assessments. The experts were teachers and master's degree students studying in the field of “Information Systems and Technologies”. Since the questions asked required the definition of one concept or explanation of one process, they were assessed as correct (0 points) and incorrect (1 point). In some cases, for example, when it was necessary to list concepts, the assessment could be expressed as a fractional value.

For clarity, here are examples of questions and some of the answers we received from students. To the question “What defines a color space?” the following answers were received: “defines the colors that are actually achievable within the chosen color model” (right); “the color representation model” (wrong); “color coordinates” (wrong); “the colors that are actually achievable within the chosen model” (right). The following answers were received for the task “Unity. Write the purpose of the Terrain object.” “By selecting GameObjec – Create Others – Terrain from the menu” (wrong); “The terrain function is a tool for creating a realistic and immersive environment for games. It allows you to create large and continuous landscapes with mountains, valleys, etc.” (right); “I don't know” (wrong); “creating a landscape” (right); “is a component for creating and editing large, open landscapes. The main purpose is to model terrain with hills, valleys, and vegetation” (right); “for drawing” (wrong); “creating a landscape” (right). Thus, the students' answers varied, could consist of one or more words, there were answers up to three sentences long.

III. METHODS

All answers were analyzed using cosine similarity [15], [16]. If a new question is added to the test, the student's answer is compared with the template answer formulated by the teacher. That is, we find the cosine value between the student's answer vector and the correct answer vector. When there are enough student answers in the database, the cosine value is calculated between the student's answer vector and the vector that is the average of the right answer vectors received from students. If the absolute value of the difference between the calculated cosine value and the expert's grade is less than 0.5, we consider that the grade obtained using the language model and the expert's grade are the same. We previously found that if the answer contains more specialized terms, a word2vec model trained on specialized texts can be used. For questions and answers that contain more common words, it is better to use large language models.

A binary classification model was investigated for questions, the answers to which were divided into 2 classes – right and wrong. In cases where the number of answers was insufficient, data augmentation was carried out using a dictionary of synonyms, chatbots based on LLM. Augmented

data (generated answers) were checked and adjusted by an expert (teacher). We also studied the capabilities of the LIME tool for interpreting the classification results. During the work, the Python programming language with the necessary libraries and the capabilities of Google Colab were used.

IV. RESULTS

Fig. 1 shows the proportion of correctly assessed answers using the BERT-based model. For different questions, this value varies from 71.4 % to 100 %. The horizontal axis shows the question numbers. Fig. 2 shows the same data grouped into intervals.

Table 1 shows the proportions of incorrectly assessed answers to the selected questions using cosine similarity. A BERT-based model was used. In this case, the value of the cosine between the vector of the student's answer and the vector that is the average of the vectors of correct answers received from the students was calculated. As can be seen from the table, the expert's assessment and the assessment obtained using cosine similarity coincided only for two questions (question 26 and question 36). Therefore, clarification is necessary and, as an additional stage of evaluation, we propose to conduct a binary classification. We considered each question separately. As stated above, the accuracy of the assessment results varies for different questions.

We propose the following steps of the algorithm for automated assessment of answers. 1). Calculate cosine similarity. 2). Prediction result using binary classification. 3). Visual interpretation of the classification for the teacher to make a final decision.

Let's look at example 1. Question “What transformations are called gradational transformations of images?” The answer is “These are transformations in which only the intensity of the corresponding point in the original image is used to calculate the intensity of a point in the resulting image” (expert rating – 1, the answer is right). Cosine similarity is 0.877. The prediction result using binary classification is 1 (the answer is right).

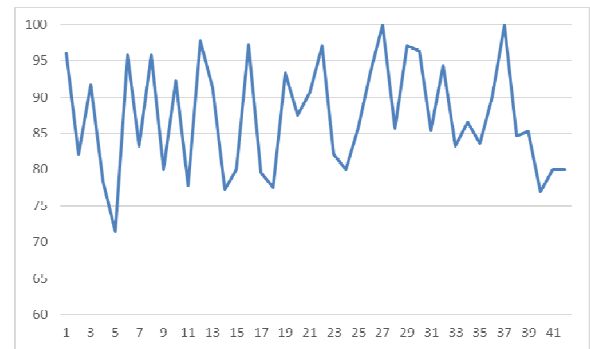


Fig. 1. The proportion of correctly assessed answers (in percent)

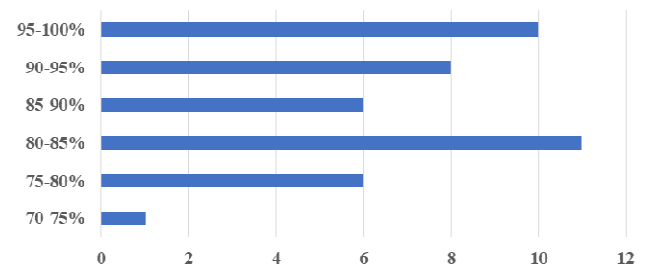


Fig. 2. Correctly assessed answers

TABLE I. PROPORTION OF INCORRECTLY ASSESSED ANSWERS

Question number	Percentage of incorrectly assessed answers	Question number	Percentage of incorrectly assessed answers
1	4	22	17.9
2	17.9	23	20
3	8.3	24	14.3
4	21.7	25	6.6
5	28.6	26	0
6	4.2	27	14.3
7	16.7	28	2.9
8	4.2	29	3.6
9	20	30	18.2
10	7.8	31	5.6
11	22.2	32	16.7
12	2.3	33	13.5
13	8.7	34	16.4
14	22.8	35	10
15	2.8	36	0
16	20.4	37	15.4
17	22.4	38	14.7
18	6.7	39	23.1
19	12.5	40	20
20	9.3	41	20
21	2.9	42	3.6

The LIME (Local Interpretable Model-Agnostic Explains) tool can be used to understand which words influenced the classification results. Here are the words with their coefficients, which show the influence of a given word on the decision (interpreting SVM decisions in text classification using TfidfVectorizer): “the” – 0.3569, “point” – 0.2499, “only” – 0.0784, “original” – 0.0649, “intensity” – 0.0635, “These” – -0.0527. The interpretation of the classification is presented in Fig. 3. Such visualizations will help the teacher to correctly evaluate the answer.

Example 2. Question “What transformations are called gradational transformations of images?” The answer is “those where only the intensity of the corresponding starting point is used” (expert rating – 1, the answer is right). Cosine similarity is 0.687. The prediction result using binary classification is 1 (the answer is right). The words with their coefficients, which show the influence of a given word on the decision: “the” – 0.2515, “point” – 0.2057, “only” – 0.1407, “those” – -0.0835, “intensity” – 0.05801, “is” – 0.0427. The interpretation of the classification is presented in Fig. 4.

Artificial intelligence tools are in development. Therefore, this interpretation can only act as an additional stage.

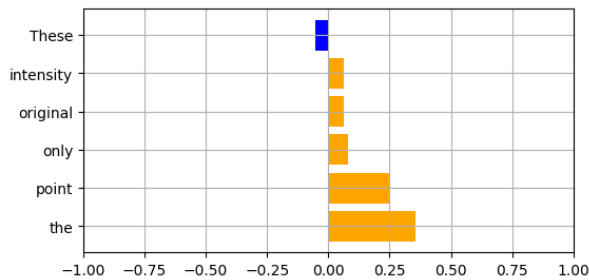


Fig. 3. Visual interpretation of the prediction for example 1

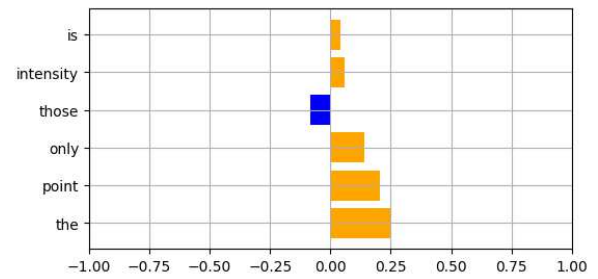


Fig. 4. Visual interpretation of the prediction for example 2

It is also possible to use LLM-based chatbots as an additional stage of assessment. Table 2 shows the results of assessing some answers obtained using ChatGPT. When accessing this chatbot, it is possible to compose a prompt in such a way as to also obtain justifications for assessments. For example, for the answer “Really achievable colors,” the explanation was: “This answer is closer to the correct one, since the color space is indeed defined by the colors that are actually achievable. However, this definition is not quite complete, since it does not take into account other factors, such as the color model and the available colors.”

The ratings determined by ChatGPT do not always coincide with the expert's ratings; they can be clarified using additional prompts. We used different prompts and various services (ChatGPT, DeepSeek). The results varied for different questions. For no question did we obtain a 100 percent match to the expert's grades. Of course, chatbots can make checking students' answers much easier. But the teacher will have to double-check these results.

It is possible to use chatbots based on large language models to formulate questions and generate answers. Generated answers can help when there is a lack of data for classification. But, as practice has shown, even in this case the teacher should double-check the results.

Note that all the examples of answers considered above are given in the form in which they were received from students. Earlier we found that the effect of correcting spelling, punctuation and other errors on the assessment result is insignificant, within the margin of error.

TABLE II. SCORING ANSWERS USING CHATGPT

Answer	Score
“Color characteristics: brightness, saturation, contrast, etc.”	0.2
“Really achievable colors.”	0.8
“Available colors in the selected model.”	0.6
“Light.”	0.3

V. CONCLUSIONS

Automated assessment of short, free-form answers remains a pressing issue. On the one hand, language models and modern natural language processing tools provide great opportunities to optimize the work of teachers. On the other hand, as the analyzes showed, erroneously estimated answers remain when using any approach. Therefore, it is necessary to apply the methods comprehensively.

The accuracy of assessing answers to open-ended questions depends on the subject area, on the volume of the filled database. It is also necessary to pay attention to the correct wording of the question. Questions that may be understood ambiguously by students should be reworded.

This paper presents an algorithm for automated assessment of short answers consisting of 1–3 sentences. The steps of the algorithm were defined as calculating cosine similarity, using binary classification models. To determine the cosine similarity, vector representations of the student's answer and the correct (template) answer are first found. The cosine value between the student's answer vector and the correct answer vector or the vector that is the average of the vectors of all correct answers in the database is calculated. Before filling the database with answers, the calculated cosine value is rounded to an integer value and presented to the teacher as a recommended grade. The teacher will double-check these grades. Once enough answers have been added to the database, the teacher will see the grade obtained using cosine similarity, prediction results and interpretation by binary classification.

We will continue to study approaches to automated assessment of answers. As the number of answers in the database increases, the accuracy of evaluation will improve. The problem is that it is impossible to obtain a large number of answers, for example, by parsing data from sites that conduct knowledge testing. Since when online testing is conducted, proctoring is not always used and students can often use the Internet, reference systems, and additional literature. Therefore, in this paper, the answers given in the classroom were analyzed.

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